MASTERS THESIS PROPOSAL

Decentralized ZKP Authorization Meshes for Industrial Robotics in DIL Environments

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A. Title of the Study

Decentralized ZKP Authorization Meshes for Industrial Robotics in DIL Environments

B. Statement of the Problem

B.1 Problem Definition

Current Industry 4.0 manufacturing systems rely on centralized cloud authentication, creating a single point of failure where network connectivity loss triggers involuntary system shutdowns. This research investigates whether decentralized Mobile Ad-hoc Network (MANET) authentication using Zero-Knowledge Proofs (ZKP) can maintain operational continuity in Disconnected, Intermittent, and Limited (DIL) network environments while quantifying the computational overhead compared to traditional cryptographic methods.

B.2 Scope and Limitations

This study focuses on industrial multi-robot systems operating under high packet loss conditions (≥20% loss rate). The research is limited to:

- Network Architecture: 10-node MANET topology with heterogeneous edge hardware
- Authentication Mechanisms: Schnorr-based Zero-Knowledge Proofs versus Elliptic Curve Cryptography (ECC-secp256k1)
- Performance Metrics: Mean Time to Recovery (MTTR), verification latency, system trust scores, and node density scalability
- Operating Environment: Simulated DIL network conditions with controlled packet loss (0%, 10%, 20%)

The study does not address post-quantum cryptographic resistance, multi-domain federation, or integration with existing enterprise identity management systems.

B.3 Research Hypotheses

H_1 (Resilience Hypothesis): Edge-First MANET authentication will reduce Mean Time to Recovery (MTTR) compared to centralized OAuth 2.0 authentication under high packet loss conditions.

H_0: There is no significant difference in MTTR between Edge-First MANET and centralized OAuth 2.0 authentication.

H_2 (Security Tax Hypothesis): Zero-Knowledge Proof (ZKP) authentication will incur higher computational overhead than Elliptic Curve Cryptography (ECC) authentication, but will provide superior trust resilience under adversarial conditions.

H_0: There is no significant difference in computational overhead or trust resilience between ZKP and ECC authentication.

H_3 (Scalability Hypothesis): Mean Time to Recovery (MTTR) will scale non-linearly with increasing node density due to verification queue saturation.

H_0: MTTR scales linearly with node density (no queue saturation effects).

B.4 Major Assumptions

1. Network Model: The MANET operates under the Dolev-Yao adversary model where attackers control the network but cannot break cryptographic primitives.
2. Hardware Reliability: Edge computing nodes operate within thermal and power specifications without hardware failures during experimental trials.
3. Packet Loss Distribution: Network disruptions follow uniform random distribution, representative of signal jamming or infrastructure degradation.
4. Trust Initialization: All legitimate nodes begin with identical initial trust scores; malicious nodes are introduced only during Byzantine fault injection testing.
5. Manufacturing Load: Peak computational load represents realistic industrial automation workloads including G-code verification, sensor fusion, and digital twin synchronization.

C. Justification for and Significance of the Study

C.1 Reason for Problem Selection

Global manufacturing experiences estimated annual losses exceeding $400 billion due to unplanned downtime, with network-related authentication failures accounting for 15-20% of incidents [1]. Current centralized authentication architectures use authorization tokens with lifetimes of 30-300 seconds; when backhaul connectivity is interrupted, tokens expire and Industrial Control Systems (ICS) trigger involuntary fail-safe shutdowns per IEC 61508 functional safety requirements [2]. This architectural dependency is incompatible with DIL network environments characteristic of contested logistics, forward-deployed military operations, and remote industrial facilities. In defense contexts, adversarial signal jamming of satellite communications renders cloud-dependent robotic systems inoperable [3],

motivating the need for decentralized authentication frameworks that decouple operational continuity from backhaul availability.

C.2 Relation to Principal Literature

Decentralized Authentication: Lin et al. (2023) demonstrated 86% authorization time reduction using blockchain-based ZK- rollups but did not test under packet loss conditions or measure end-to-end recovery times [4]. Burgos et al. (2024) proposed decentralized IoT authentication with signature aggregation but focused on bandwidth efficiency rather than resilience metrics [5]. No existing papers measure MTTR in DIL/MANET environments with ≥20% packet loss, establishing the novelty of H_1.

Cryptography on Constrained Hardware: Chatzigiannakis et al. (2011) implemented ECC-based ZK interactive proofs on microcontrollers, providing execution time benchmarks for constrained devices [6]. However, no papers provide direct ZKP versus ECC comparative performance data on ARM Cortex-A or Cortex-M4 platforms under realistic Manufacturing Execution System computational load. The processing ceiling concept—maximum tolerable verification latency before MTTR exceeds operational thresholds—has not been quantified in prior literature, establishing the novelty of H_2.

Scalability in Authenticated MANET Systems: Barrion et al. (2024) demonstrated dynamic trust propagation in robotic swarms at n=8 nodes but did not implement cryptographic verification or measure MTTR under adversarial conditions [7]. Lv et al. (2020) explored decentralized blockchain architectures but did not analyze verification queuing or livelock emergence [8]. No papers quantify livelock thresholds in ZKP-authenticated MANET systems at n=10 operational density, establishing the novelty of H_3.

C.3 Importance and Significance

Advancement of Knowledge: This research provides the first empirical characterization of the resilience-security-scalability trade-off space for decentralized authentication in industrial multi-robot systems. By quantifying the processing ceiling where cryptographic verification overhead causes queue saturation, the study establishes design parameters for edge-first authentication architectures operating under contested network conditions.

Engineering Impact: The proposed Edge-First MANET framework addresses IEC 61508 functional safety requirements by maintaining operational continuity during backhaul connectivity loss, preventing involuntary safety stops that halt manufacturing operations. Quantifying MTTR reduction compared to centralized baselines provides actionable design criteria for expeditionary manufacturing, forward-deployed military logistics, and remote industrial facilities where cloud connectivity cannot be guaranteed.

Defense Applications: Current DoD robotic platforms require continuous operator control or cloud-based mission planning, becoming inert during communications blackout [3]. The research directly supports DoD requirements for contested logistics by enabling continued operation under adversarial signal jamming while quantifying computational overhead and scalability limits on resource-constrained edge hardware.

Significance to the Student: Building on the candidate's foundational research in SWARM-SLAM robotics and tactical C5 infrastructure automation at USMC MCTSSA, this thesis serves as the R&D engine for a Dual-Path trajectory in defense entrepreneurship and military cyber operations. By mastering Zero-Knowledge Proofs on edge hardware and MANET protocol design, the work directly targets DoD "Contested Logistics." This progression from undergraduate swarm modeling and Marine Corps automation to resilient, decentralized command and control establishes the advanced cyber-physical expertise required for both deep-tech commercialization and future battlespace command.

D. Methodology

D.1 Research Design

This study employs a comparative experimental design using a 2×2×3 mixed factorial structure. Between-subjects factors are Authentication Mechanism (ZKP vs. ECC) and Network Condition (Baseline vs. 20% Packet Loss). The within-subjects factor is Node Density (n=3, n=6, n=10). Each experimental condition will be replicated 30 times to achieve statistical power ≥0.80 for detecting medium effect sizes (Cohen's d=0.8) at α=0.05 significance level [9].

D.2 Study Population and Setting

The study population consists of a 10-node heterogeneous MANET testbed representing a simplified industrial manufacturing cell: one supervisor node running Manufacturing Execution System logic, five worker nodes executing equipment control, three sensor nodes for material verification and environmental monitoring, and one robot node (6-axis collaborative robot). All experiments will be conducted in an isolated laboratory environment with dedicated network infrastructure.

D.3 Data Collection Schedule

1. Baseline Characterization: Measure centralized OAuth 2.0 authentication performance under varying packet loss conditions.
2. ZKP Implementation: Deploy Schnorr-based Zero-Knowledge Proof authentication and measure verification latency under idle and peak computational load.
3. ECC Baseline Implementation: Implement ECDSA (secp256k1) authentication for direct comparison.

4. Byzantine Fault Injection: Introduce adversarial nodes broadcasting forged credentials and measure system trust score degradation.
5. Scalability Testing: Conduct node density sweeps to measure MTTR scaling, gossip protocol overhead, and livelock emergence.
6. Statistical Analysis: Execute normality testing, mixed factorial ANOVA, post-hoc comparisons, and nonlinear regression.

D.4 Data Collection Instruments

Primary Instrumentation: ROS2 bag files capturing all MQTT messages, DDS traffic, and cryptographic operations with microsecond-precision timestamps; network traffic captures via tcpdump; CPU utilization and thermal monitoring.

Measurement Validity: Hardware-synchronized clocks prevent drift across distributed nodes. Verification latency measurements exclude network transmission time by instrumenting cryptographic library function calls directly. MTTR measurements are decomposed into constituent phases (detection, authentication, recovery).

Measurement Reliability: Each experimental condition is replicated 30 times with randomized trial ordering (Latin Square design) to control for learning effects. Outliers exceeding three standard deviations are excluded using Grubbs' test ($\alpha=0.05$). Data analysts are blinded to authentication mechanism conditions during preprocessing.

D.5 Key Terms and Concepts

Mean Time to Recovery (MTTR): Elapsed time from network partition detection to full restoration of authenticated communication, measured in milliseconds.

Disconnected, Intermittent, and Limited (DIL) Networks: Network environments characterized by frequent connectivity interruptions, high packet loss rates (≥20%), and limited bandwidth.

Processing Ceiling: Maximum sustainable cryptographic verification throughput before queue saturation causes MTTR to exceed operational thresholds, resulting in Head-of-Line blocking.

Probabilistic Trust Score (Γ): Continuous metric $\Gamma(t) \in [0,1]$ updated via exponential weighted moving average based on cryptographic verification outcomes, enabling graceful degradation under intermittent connectivity.

Livelock: System state where nodes continuously attempt cryptographic verification but make no forward progress due to gossip protocol traffic saturating MANET bandwidth.

D.6 Data Processing and Analysis Procedures

Statistical Analysis Plan:

1. Normality Testing: Shapiro-Wilk test on MTTR distributions ($\alpha=0.05$). If normality assumptions are violated, non-parametric alternatives (Kruskal-Wallis) will be employed.
2. H_1 Validation: Two-way ANOVA (Authentication Mechanism × Packet Loss) with MTTR as dependent variable. Post-hoc Tukey HSD tests for pairwise comparisons. Effect sizes calculated using Cohen's d.
3. H_2 Validation: Paired t-test comparing verification latency (t_verify_ZKP vs. t_verify_ECC) under load. Independent t-test comparing system trust scores (Γ_ZKP vs. Γ_ECC) under Byzantine fault injection. Significance threshold $p<0.05$ with Bonferroni correction.

4. H_3 Validation: Nonlinear regression fitting $MTTR(n) = k \cdot n^{\alpha}$ to empirical data across node densities. Hypothesis validation requires $\alpha>1$ (super-linear scaling) with $p<0.05$ significance.
5. Power Analysis: Post-hoc verification that achieved statistical power ≥ 0.80 for all hypothesis tests.
6. Threat Mitigation: Randomized trial ordering, counterbalancing of authentication mechanism presentation, dedicated VLAN isolation. Hardware platform selection representative of industrial edge computing deployments. Packet loss rates consistent with contested logistics environments. MTTR operationalization validated against IEC 61508 functional safety standards [2].

E. Resources Required

E.1 Hardware Resources

- 10-Node MANET Testbed: 1× Raspberry Pi 4 (8GB RAM, 64GB storage) supervisor, 5× Raspberry Pi 4 (4GB RAM) workers, 3× Arduino Nano 33 BLE (Cortex-M4, 256KB SRAM) sensors, 1× Niryo Ned2 6-axis robot
- Network Infrastructure: Gigabit Ethernet switch, IEEE 802.11ac wireless access points, packet loss injection tools (Linux Traffic Control, NetEm)
- Laboratory Facilities: URI Robotics Lab with isolated testbed environment

Accessibility: All hardware resources are currently procured and operational in the URI Robotics Lab under Dr. Sodhi's supervision. The testbed has been validated for basic functionality and is available for exclusive use during the experimental phase.

E.2 Software Resources

- Operating System and Middleware: Ubuntu 22.04 LTS, ROS2 Humble Hawksbill, Eclipse Cyclone DDS
- Cryptographic Libraries: libsecp256k1 (Bitcoin Core), libsodium 1.0.x
- Data Analysis Tools: Python 3.10+ with NumPy, SciPy, Pandas, Matplotlib
- Literature Access: IEEE Xplore, ACM Digital Library, arXiv, URI Library interlibrary loan

Accessibility: All software resources are open-source or available through URI site licenses. The candidate has demonstrated proficiency in ROS2, Python, and cryptographic library integration through prior coursework and research projects.

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